Appendix: Matroid Theory Background

This appendix collects the matroid theory used in the CS3263 report and its classification paper. All matroids are finite, and $\mathbb{F}$ denotes an arbitrary field. See Oxley for a fuller treatment and proofs of the standard results [1, 2].

1 Basic definitions

Definition 1.1. A matroid M is an ordered pair $(E, \mathcal{I})$ consisting of a finite set E and a collection $\mathcal{I}$ of subsets of E having the following three properties:

1. $\emptyset \in \mathcal{I}$.
2. If $I \in \mathcal{I}$ and $I' \subseteq I$, then $I' \in \mathcal{I}$.
3. If I_1 and I_2 are in $\mathcal{I}$ and $|I_1| < |I_2|$, then there exists $e \in I_2 - I_1$ such that $I_1 \cup \{e\} \in \mathcal{I}$.

The members of $\mathcal{I}$ are the independent sets of M, and E is its ground set. A subset of E outside $\mathcal{I}$ is dependent. We often write $\mathcal{I}(M)$ for $\mathcal{I}$ and $E(M)$ for E.

Definition 1.2. Let E be the set of column labels of an $m \times n$ matrix A over a field $\mathbb{F}$, and let $\mathcal{I}$ consist of the subsets $X \subseteq E$ for which the columns indexed by X are linearly independent. Then $(E, \mathcal{I})$ is the vector matroid of A, denoted by $M[A]$.

It is easy to check that $M[A]$ is indeed a matroid.

Definition 1.3. Two matroids M_1 and M_2 are isomorphic, written $M_1 \cong M_2$, if there is a bijection ψ from $E(M_1)$ to $E(M_2)$ such that, for all $X \subseteq E(M_1)$, the set $\psi(X)$ is independent in M_2 if and only if X is independent in M_1.

Definition 1.4. If a matroid M is isomorphic to the vector matroid of a matrix A over a field $\mathbb{F}$, then M is $\mathbb{F}$-representable. The matrix A is an $\mathbb{F}$-representation of M.

2 Circuits

Definition 2.1. A minimal dependent set in a matroid M is called a circuit. We denote the set of circuits of M by $\mathcal{C}$ or $\mathcal{C}(M)$. A circuit with n elements is called an n-circuit.

A matroid is uniquely determined by its set of circuits $\mathcal{C}$: the members of $\mathcal{I}$ are exactly those subsets of E that contain no member of $\mathcal{C}$.

Lemma 2.2. *The set $\mathcal{C}$ of circuits of a matroid has the following properties:*

1. *$\emptyset \notin \mathcal{C}$.*
2. *If C_1 and C_2 are members of $\mathcal{C}$ and $C_1 \subseteq C_2$, then $C_1 = C_2$.*
3. *If C_1 and C_2 are distinct members of $\mathcal{C}$ and $e \in C_1 \cap C_2$, then there is a member C_3 of $\mathcal{C}$ such that $C_3 \subseteq (C_1 \cup C_2) \setminus \{e\}$.*

It turns out that the converse also holds in the sense that we can define a matroid using the above properties:

Theorem 2.3. *Let E be a set and $\mathcal{C}$ be a collection of subsets of E satisfying the three conditions above. Let $\mathcal{I}$ be the collection of subsets of E that contain no member of $\mathcal{C}$. Then $(E, \mathcal{I})$ is a matroid having $\mathcal{C}$ as its collection of circuits.*

Lemma 2.4. *Suppose I is an independent set in a matroid M and $e \in E(M) \setminus I$ is such that $I \cup \{e\}$ is dependent. Then M has a unique circuit contained in $I \cup \{e\}$, and this circuit contains e.*

A cycle of a graph is a nonempty connected subgraph in which each vertex has degree two, counting a loop twice toward the degree.

Definition 2.5. Let E be the set of edges of a graph G and $\mathcal{C}$ be the set of edge sets of cycles of G. Then $\mathcal{C}$ is a set of circuits of a matroid on E called the cycle matroid or polygon matroid of G. It is denoted by $M(G)$.

In other words, X is an independent set of $M(G)$ if and only if the subgraph induced by X is a forest. One can check that the set $\mathcal{C}$ indeed satisfies the three properties for a set of circuits. A matroid is graphic if it is isomorphic to the cycle matroid of a graph.

Definition 2.6. We record some graph inspired terminologies for general matroid:

1. An element e of a matroid M is called a loop if $\{e\}$ is a circuit of M.
2. Distinct elements f and g are parallel if $\{f, g\}$ is a circuit.
3. A parallel class of M is a maximal nonempty set of nonloops whose distinct elements are pairwise parallel. A parallel class is trivial if it contains just one element.
4. If a matroid has no loops and no non-trivial parallel classes, it is called a simple matroid.

Every basis avoids loops. If e_1 is a nonloop and e_2 is distinct from e_1, then the two elements are parallel exactly when replacing e_1 by e_2 in any basis containing e_1 gives another basis. In a parallel class X, write $\mathcal{B}_x$ for the bases containing $x \in X$. For distinct $x, y \in X$, the families $\mathcal{B}_x$ and $\mathcal{B}_y$ are disjoint, and the exchange property for parallel elements gives

$$\{B \setminus \{x\} : B \in \mathcal{B}_x\} = \{B \setminus \{y\} : B \in \mathcal{B}_y\}.$$

The *simplification* $\mathrm{si}(M)$ is obtained by deleting every loop and retaining one element from each parallel class. Different choices give isomorphic matroids.

3 Bases

Definition 3.1. A maximal independent set in M is called a basis or a base of M.

Lemma 3.2. *All bases of a matroid have the same cardinality.*

Lemma 3.3 (Basis exchange)**.** *Let $\mathcal{B}$ be the bases of a matroid. If $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \setminus B_2$, then some $y \in B_2 \setminus B_1$ satisfies $(B_1 \setminus \{x\}) \cup \{y\} \in \mathcal{B}$.*

Theorem 3.4. *Let $\mathcal{B}$ be a nonempty family of equicardinal subsets of a finite set E, satisfying the exchange property above. The subsets of members of $\mathcal{B}$ form the independent sets of a matroid whose bases are exactly $\mathcal{B}$.*

Lemma 3.5. *Let B be a basis of M. If $e \in E(M) \setminus B$, then $B \cup \{e\}$ contains a unique circuit $C(e, B)$, which contains e.*

We call $C(e, B)$ the fundamental circuit of e with respect to B.

Proposition 3.6. *Every circuit in a matroid is the fundamental circuit of some element with respect to some basis.*

4 Rank

Let $M = (E, \mathcal{I})$ and $X \subseteq E$. The family $\mathcal{I}|X = \{I \subseteq X : I \in \mathcal{I}\}$ defines a matroid on X, called the restriction $M|X$ or the deletion $M \setminus (E \setminus X)$. Its circuits are exactly the circuits of M contained in X.

Matroid deletion is a direct generalisation of the corresponding concept for graphs. For a graph G and a subset T of $E(G)$, recall that $G\backslash T$ is the graph obtained from G by deleting the edges in T. We have

$$M(G\backslash T) = M(G)\backslash T.$$

Definition 4.1. We define the rank $r(X)$ of X to be the cardinality of a basis B of $M|X$ and call such a set B a basis of X.

Lemma 4.2. *The rank function r has the following properties:*

1. *If $X \subseteq E$, then $0 \leq r(X) \leq |X|$.*
2. *If $X \subseteq Y \subseteq E$, then $r(X) \leq r(Y)$.*
3. *If X and Y are subsets of E, then*

$$r(X \cup Y) + r(X \cap Y) \leq r(X) + r(Y).$$

These properties characterise the rank function and in fact the matroid.

Theorem 4.3. *Let E be finite and $r\colon 2^E \to \mathbb{Z}_{\geq 0}$ satisfy the three properties above. Then $\mathcal{I} = \{X \subseteq E : r(X) = |X|\}$ defines a matroid on E whose rank function is r.*

Proposition 4.4. *Let M be a matroid with rank function r and let $X \subseteq E(M)$. Then*

1. *X is independent if and only if $|X| = r(X)$*
2. *X is a basis if and only if $|X| = r(X) = r(M)$*
3. *X is a circuit if and only if X is nonempty and, for all $x \in X$, $r(X - \{x\}) = |X| - 1 = r(X)$.*

5 Closure

Let M be a matroid on E with rank function r.

Definition 5.1. Let $\mathrm{cl}\colon 2^E \to 2^E$ be defined by

$$\mathrm{cl}(X) = \{x \in E : r(X \cup \{x\}) = r(X)\}.$$

This function is called the closure operator of M, and we call $\mathrm{cl}(X)$ the closure or span of X in M.

Lemma 5.2. *For all X, we have $r(X) = r(\mathrm{cl}(X))$.*

Lemma 5.3. *The closure operator of a matroid on a set E has the following properties:*

1. *If $X \subseteq E$, then $X \subseteq \mathrm{cl}(X)$.*
2. *If $X \subseteq Y \subseteq E$, then $\mathrm{cl}(X) \subseteq \mathrm{cl}(Y)$.*
3. *If $X \subseteq E$, then $\mathrm{cl}(\mathrm{cl}(X)) = \mathrm{cl}(X)$.*
4. *If $X \subseteq E$, $x, y \in E$, and $y \in \mathrm{cl}(X \cup \{x\}) \setminus \mathrm{cl}(X)$, then $x \in \mathrm{cl}(X \cup \{y\})$.*

The fourth property is sometimes referred to as the MacLane-Steinitz exchange property.
Once again we have a characterisation of matroids in terms of such functions.

Theorem 5.4. *Let E be a set and* $\mathrm{cl}\colon 2^E \to 2^E$ *be a function satisfying the above four properties. Let*

$$\mathcal{I} = \{X \subseteq E : x \notin \mathrm{cl}(X \setminus \{x\}) \text{ for all } x \in X\}.$$

Then $(E, \mathcal{I})$ is a matroid having closure operator cl.

Definition 5.5. A subset $X \subseteq E(M)$ for which $\mathrm{cl}(X) = X$ is a flat or closed set of M. A hyperplane of M is a flat of rank $r(M) - 1$.

A subset X is a spanning set of M if $\mathrm{cl}(X) = E(M)$. It spans a subset Y of $E(M)$ if $Y \subseteq \mathrm{cl}(X)$.

Proposition 5.6. *The subset X has the following properties:*

1. *X is a spanning set if and only if $r(X) = r(M)$.*
2. *X is a basis if and only if it is both spanning and independent.*
3. *X is a basis if and only if it is a minimal spanning set.*
4. *X is a hyperplane if and only if it is a maximal non-spanning set.*

Proposition 5.7.

$$\mathrm{cl}(X) = X \cup \{x \in E(M) : M \text{ has a circuit } C \text{ with } x \in C \subseteq X \cup \{x\}\}.$$

6 Duality

Theorem 6.1. *Let M be a matroid and $\mathcal{B}^*(M)$ be $\{E(M) - B : B \in \mathcal{B}(M)\}$. Then $\mathcal{B}^*(M)$ is the set of bases of a matroid on $E(M)$.*

Definition 6.2. The matroid defined in the last theorem is called the dual of M, denoted as M^*.

By construction, $(M^*)^* = M$. The bases of M^* are the cobases of M, and its circuits are the cocircuits of M. A *coloop* of M is an element in every basis of M, or equivalently a loop of M^*.

Proposition 6.3. *Let M be a matroid on E and let $X \subseteq E$. Then*

1. *X is independent in M if and only if $E \setminus X$ spans M^*.*
2. *X spans M if and only if $E \setminus X$ is independent in M^*.*
3. *X is a hyperplane of M if and only if $E \setminus X$ is a cocircuit of M.*
4. *X is a circuit of M if and only if $E \setminus X$ is a hyperplane of M^*.*

In general, we attach an asterisk to a symbol to denote association with the dual. For example, r^* denotes the rank function of M^* while $\mathcal{C}^*$ denotes its set of circuits.

Proposition 6.4. *For all $X \subseteq E(M)$,*

$$r^*(X) = r(E - X) + |X| - r(M).$$

Proposition 6.5. *In a matroid M, let C be a circuit and C^* be a cocircuit. Then $|C \cap C^*| \neq 1$.*

The above property is often called orthogonality.

Theorem 6.6. *If a matroid M is $\mathbb{F}$-representable, then M^* is also $\mathbb{F}$-representable.*

7 Minors

Let M be a matroid on E and $T \subseteq E$.

Definition 7.1. The contraction of T in M is $M/T = (M^* \setminus T)^*$.

Evidently M/T has ground set $E - T$. We record a series of facts about the distinguished subsets, rank and closure of a contraction.

Proposition 7.2. *For all $X \subseteq E \setminus T$,*

$$r_{M/T}(X) = r_M(X \cup T) - r_M(T).$$

Proposition 7.3.

$$\mathcal{I}(M/T) = \{I \subseteq E \setminus T : M|T \text{ has a basis } B \text{ with } I \cup B \in \mathcal{I}(M)\}.$$

Corollary 7.4.

$$\mathcal{B}(M/T) = \{B' \subseteq E \setminus T : M|T \text{ has a basis } B \text{ with } B' \cup B \in \mathcal{B}(M)\}.$$

Proposition 7.5. *The circuits of M/T consist of the minimal nonempty members of $\{C - T : C \in \mathcal{C}(M)\}$.*

Proposition 7.6. *For all $X \subseteq E \setminus T$,*

$$\mathrm{cl}_{M/T}(X) = \mathrm{cl}_M(X \cup T) - T.$$

$M \setminus T$ and M/T have the same ground set, and

$$\mathcal{I}(M/T) \subseteq \mathcal{I}(M \setminus T).$$

Proposition 7.7. *$M \setminus T = M/T$ if and only if $r(T) + r(E - T) = r(M)$.*

Proposition 7.8. *In a matroid M, let T_1 and T_2 be disjoint subsets of $E(M)$. Then*

1. $(M \setminus T_1) \setminus T_2 = M \setminus (T_1 \cup T_2) = (M \setminus T_2) \setminus T_1$
2. $(M/T_1)/T_2 = M/(T_1 \cup T_2) = (M/T_2)/T_1$
3. $(M \setminus T_1)/T_2 = (M/T_2) \setminus T_1$.

Definition 7.9. A minor of M is a matroid of the form $(M \setminus X)/Y$ for disjoint subsets X, Y of $E(M)$. It is a proper minor when $X \cup Y$ is nonempty.

The commutation identities put any sequence of deletions and contractions in this form. An N-minor of M is a minor of M isomorphic to N.

Proposition 7.10. *A matroid N is a minor of a matroid M if and only if N^* is a minor of M^*. More particularly,*

$$N = M \backslash X / Y \iff N^* = M^* / X \backslash Y.$$

Proposition 7.11. *Let M_1 and M_2 be matroids on a common ground set E and suppose $e \in E$. Then the following statements are equivalent:*

1. *$M_1 \backslash \{e\} = M_2 \backslash \{e\}$ and $M_1/\{e\} = M_2/\{e\}$*
2. *$M_1 = M_2$, or their deletions of e agree and e is a loop of one matroid and a coloop of the other.*

Graph contraction deletes a nonloop edge and identifies its ends. Contraction of a loop means deleting it. For $T \subseteq E(G)$, G/T is obtained by contracting the edges in T in any order.

Proposition 7.12. *If G is a graph, then $M(G)/T = M(G/T)$ for all $T \subseteq E(G)$.*

Corollary 7.13. *Every minor of a graphic matroid is graphic.*

For a matrix A, we denote by $A\backslash T$ the matrix obtained from A by deleting all the columns whose labels are in T. Evidently

$$M[A]\backslash T = M[A\backslash T].$$

The duality and deletion results give closure under minors.

Proposition 7.14. *Every minor of an $\mathbb{F}$-representable matroid is $\mathbb{F}$-representable.*

Definition 7.15. An excluded minor for $\mathbb{F}$-representability is a nonrepresentable matroid over $\mathbb{F}$ whose proper minors are all $\mathbb{F}$-representable.

8 Connectivity

Proposition 8.1. *Every graphic matroid has a connected graph representation. To obtain one, take a graph representing the matroid and identify one vertex from each connected component. Every cycle is contained in a single original component, so the cycle matroid is preserved.*

A vertex cut of a graph G is a set of vertices whose removal increases the number of connected components. A one-vertex cut is called a cut vertex. For a connected noncomplete graph, the vertex connectivity $\kappa(G)$ is the minimum size of a vertex cut. For a complete graph $\kappa(G) = |V(G)| - 1$, and for a disconnected graph $\kappa(G) = 0$. A graph is k-connected if $\kappa(G) \geq k$.

Lemma 8.2. *Let G be a connected loopless graph with at least three vertices. Then G is 2-connected if and only if every pair of distinct edges lies on a common cycle.*

The connectedness hypothesis matters because a cycle matroid does not record isolated vertices or distinguish separate components joined at a single vertex.

Definition 8.3. A matroid M is connected, or 2-connected, if every pair of distinct elements of $E(M)$ lies on a common circuit.

Proposition 8.4. *A matroid M is connected if and only if M^* is connected.*

The components of M are the equivalence classes of the relation in which two elements coincide or lie on a common circuit. A separator is a union of components. The sets $\emptyset$ and $E(M)$ are the trivial separators, and the complement of any separator is also a separator.

Proposition 8.5. *A subset $X \subseteq E(M)$ is a separator if and only if*

$$r_M(X) + r_M(E(M) \setminus X) = r(M).$$

Definition 8.6. Let M_1 and M_2 be matroids on disjoint sets E_1 and E_2. Their direct sum or 1-sum $M_1 \oplus M_2$ has ground set $E_1 \cup E_2$ and independent sets

$$\{I_1 \cup I_2 : I_1 \in \mathcal{I}(M_1),\ I_2 \in \mathcal{I}(M_2)\}.$$

Proposition 8.7. *A matroid M is disconnected if and only if there is a proper nonempty subset $T \subset E(M)$ such that*

$$M = (M|T) \oplus (M|(E(M) \setminus T)).$$

Definition 8.8. Let M and N each have at least three elements, with $E(M) \cap E(N) = \{p\}$. Suppose p is neither a loop nor a coloop of either matroid. The 2-sum $M \oplus_2 N$ has ground set $(E(M) \cup E(N)) \setminus \{p\}$. Its circuits are those of $M \setminus p$ and $N \setminus p$, together with the sets $(C \cup D) \setminus \{p\}$ where $C \in \mathcal{C}(M)$ and $D \in \mathcal{C}(N)$ both contain p.

Proposition 8.9. *The classes of graphic and $\mathbb{F}$-representable matroids are closed under direct sums and 2-sums.*

Matroid connectivity can be expressed in terms of rank.

Definition 8.10. For $X \subseteq E(M)$, the connectivity function of M is

$$\lambda_M(X) = r_M(X) + r_M(E(M) \setminus X) - r(M).$$

The dual rank formula gives $\lambda_M(X) = r_M(X) + r_{M^*}(X) - |X|$.

Definition 8.11. For a positive integer k, a set $X \subseteq E(M)$ is k-separating if $\lambda_M(X) < k$. The partition $(X, E(M) \setminus X)$ is a k-separation if X is k-separating and both sides have at least k elements. The matroid M is n-connected if it has no k-separation for $1 \leq k < n$.

A set X is a separator exactly when $\lambda_M(X) = 0$. The nontrivial separators give the 1-separations.

Corollary 8.12. *For every $X \subseteq E(M)$, $\lambda_M(X) = \lambda_{M^*}(X)$. Hence M is n-connected if and only if M^* is n-connected.*

The following decomposition theorem is used in the classification [2, Section 8.3].

Theorem 8.13. *A connected matroid M has a 2-separation if and only if it can be expressed as a 2-sum $M_1 \oplus_2 M_2$ in which each summand has at least three elements and is isomorphic to a proper minor of M.*

References

[1] J. Oxley, *What is a matroid?*, revised survey. `https://www.math.lsu.edu/~oxley/survey4.pdf`

[2] J. Oxley, *Matroid Theory*, 2nd ed., Oxford University Press, 2011.